WORKED SOLUTION

STEP II 1987, Question 2

Pure • Trigonometric Identities

STEP II 1987, Question 2

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/87-s2-q2

The key idea

The expression only involves $\sin^2 A$, $\sin^2 B$ and $\sin^2 C$, and it is a *quadratic* in $\sin^2 C$. The condition on $A \pm B \pm C$ says that C is tied to $A + B$ or $A - B$, so it is natural to guess that the two roots of that quadratic are $\sin^2(A + B)$ and $\sin^2(A - B)$. Checking the sum and product of the roots confirms the guess, and the result drops out.

Solution

Write $x = \sin^2 A$, $y = \sin^2 B$, $z = \sin^2 C$, and call the left-hand side E. Then

$$E = x^2 + y^2 + z^2 - 2yz - 2zx - 2xy + 4xyz.$$

Collecting powers of z,

$$\begin{aligned} E &= z^2 - 2z(x + y - 2xy) + (x^2 - 2xy + y^2) \\ &= z^2 - 2(x + y - 2xy)\,z + (x - y)^2. \end{aligned}$$

The roots of the quadratic

We claim that

$$E = (z - \sin^2(A + B))(z - \sin^2(A - B)).$$

Since both sides are monic quadratics in z, it is enough to check that $\sin^2(A + B)$ and $\sin^2(A - B)$ have the right sum and product.

Sum. Expanding $\sin(A \pm B) = \sin A \cos B \pm \cos A \sin B$ and squaring, the cross terms cancel when we add:

$$\begin{aligned} \sin^2(A + B) + \sin^2(A - B) &= 2\sin^2 A \cos^2 B + 2\cos^2 A \sin^2 B \\ &= 2x(1 - y) + 2(1 - x)y \\ &= 2(x + y - 2xy). \end{aligned}$$

Product. Using the difference of two squares,

$$\sin(A+B)\sin(A-B) = \sin^2 A\cos^2 B - \cos^2 A\sin^2 B = x(1-y) - (1-x)y = x - y,$$

so $\sin^2(A+B)\sin^2(A-B) = (x-y)^2$.

Both match the coefficients of E, which proves the claim.

> **Ask yourself** Why is it enough to know that $\sin^2 C$ equals $\sin^2(A+B)$ or $\sin^2(A-B)$?

Because then z is one of the two roots, so one factor of E vanishes. It remains to show that the condition on the angles gives exactly this.

Using the condition

Suppose $A + \varepsilon_1 B + \varepsilon_2 C = k\pi$ for some integer k and some signs $\varepsilon_1, \varepsilon_2 \in \{1, -1\}$. Then

$$C = \varepsilon_2(k\pi - (A + \varepsilon_1 B)).$$

The function $\sin^2$ is even and has period π, so neither the sign ε_2 nor the $k\pi$ changes its value:

$$\sin^2 C = \sin^2(A + \varepsilon_1 B).$$

So $z = \sin^2(A+B)$ (when $\varepsilon_1 = 1$) or $z = \sin^2(A-B)$ (when $\varepsilon_1 = -1$). In either case one factor of E is zero, and therefore

$$E = 0,$$

as required. ■

A neater form (and the converse)

Each factor is a difference of two squares of sines, and $\sin^2 P - \sin^2 Q = \sin(P+Q)\sin(P-Q)$ (the product formula above with $A = P$, $B = Q$). So

$$E = \sin(A+B+C)\,\sin(C-A-B)\,\sin(C+A-B)\,\sin(C-A+B).$$

The four angles here are, up to sign, exactly $A+B+C$, $A+B-C$, $A-B+C$ and $A-B-C$. This makes the result transparent: $E = 0$ *if and only if* one of the four angles $A \pm B \pm C$ is a multiple of π.

Checks

- Take $A = \frac{\pi}{6}$, $B = \frac{\pi}{3}$, $C = \frac{\pi}{2}$, so $A + B - C = 0$. Then $x = \frac{1}{4}$, $y = \frac{3}{4}$, $z = 1$ and $E = \frac{1}{16} + \frac{9}{16} + 1 - \frac{3}{2} - \frac{1}{2} - \frac{3}{8} + \frac{3}{4} = 0$.
- With $A = 0.3$, $B = 0.7$, $C = 1.1$ (no special relation), both E and the four-sine product evaluate to 0.05538, so the expression is genuinely non-zero away from the condition.

- A random test over 10^5 triples confirmed the factorisation, and that $E = 0$ to within 10^{-14} whenever one of $A \pm B \pm C$ is forced to be a multiple of π.

Common slips

- Treating only the case $A + B + C = \pi$ (a triangle). The question allows any of the four sign patterns and any multiple of π; the evenness and π-periodicity of $\sin^2$ handle them all at once.
- Solving the quadratic with the formula and getting tangled in $\sqrt{xy(1-x)(1-y)}$. Checking the sum and product of the guessed roots avoids square roots and sign worries entirely.

Beyond STEP

The expression $x^2 + y^2 + z^2 - 2yz - 2zx - 2xy$ appears all over mathematics: with x, y, z replaced by a^2, b^2, c^2 it is -16 times the square of the area of a triangle with sides a, b, c (Heron's formula), and it is also the discriminant-like quantity that decides whether three lengths can form a triangle. Seeing a symmetric polynomial as a quadratic in one variable, and guessing its roots, is a technique that recurs throughout algebra and invariant theory. For more problems in this spirit, see Siklos, *Advanced Problems in Mathematics*.

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